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Problem Set 8

Mathematics for Social Scientists

EXERCISE 1. *(Essential Mathematics for Economic Analysis (5th edition),* Knut Sydsaeter and Peter Hammond, Section 15.3, Question 5)

Consider the three matrices $\mathbf{A} = \begin{pmatrix} 2 & 2 \\ 1 & 5 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 3 & 2 \end{pmatrix}$ and $\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(i) Find a matrix $\mathbf{C}$ such that $(\mathbf{A} - 2\mathbf{I})\mathbf{C} = \mathbf{I}$.

(ii) Is there a matrix $\mathbf{D}$ satisfying $(\mathbf{B} - 2\mathbf{I})\mathbf{D} = \mathbf{I}$?

EXERCISE 2. Determine the rank of the following matrices:

(i)

$$A = \begin{pmatrix} 3 & 7 \\ -7 & 3 \end{pmatrix}$$

(ii)

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \end{pmatrix}$$

EXERCISE 3. Compute the determinant of the following matrices. If they are invertible, find the inverse.

$$C = \begin{pmatrix} 2 & 7 \\ 7 & 8 \end{pmatrix}, \qquad D = \begin{pmatrix} x & x^2 \\ x^3 & x^4 \end{pmatrix}, \qquad E = \begin{pmatrix} -3 & 0 & 4 \\ 1 & 4 & -2 \\ 7 & -6 & 3 \end{pmatrix}.$$

EXERCISE 4. Does the property $\det(A+B) = \det(A) + \det(B)$ hold for $A, B \in \mathbb{R}^{n \times n}$ and any $n \geq 1$?

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Exercise 5. Compute the determinant of the following matrices:

$$A = \begin{pmatrix} 2 & -3 & 2 \\ 1 & 4 & 0 \\ 0 & 1 & -5 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 & 2 \\ 4 & 0 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

Are they invertible?

Exercise 6. Find eigenvalues and eigenvectors of the matrix

$$B = \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}.$$

Exercise 7. Compute the characteristic polynomial $P_C(\lambda)$ for $C = \begin{pmatrix} 0 & -1 & 1 \\ -3 & -2 & 3 \\ -2 & -2 & 3 \end{pmatrix}$ and try to find the eigenvalues!

Exercise 8. Does the matrix $D = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ have (real-valued) eigenvalues?